

The Modern Computer Vision Pipeline

Technical Research Report

Week 1 Assignment

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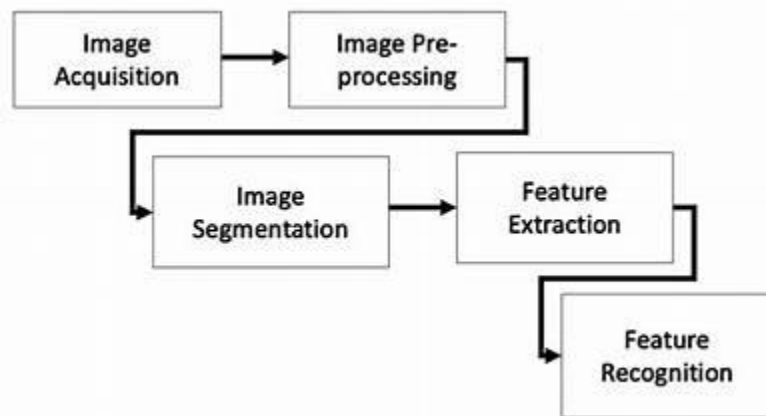
Computer Vision Engineering Track

Introduction

Computer Vision is a branch of Artificial Intelligence (AI) that enables computers to interpret and understand images and videos in a way similar to human vision. It combines image processing, pattern recognition, and machine learning techniques to extract useful information from visual data.

Today, Computer Vision is used in many industries including healthcare, autonomous vehicles, manufacturing, agriculture, security surveillance, robotics, retail, and smart cities. Applications such as facial recognition, object detection, medical image analysis, traffic monitoring, and quality inspection rely heavily on Computer Vision technologies.

A modern Computer Vision system follows a structured pipeline. Each stage performs a specific task that transforms raw image data into meaningful information for decision-making. Understanding this pipeline is essential for developing reliable Computer Vision applications.



1. Image Acquisition

Image acquisition is the first stage of the Computer Vision pipeline. It involves capturing images or videos using cameras, webcams, drones, mobile phones, CCTV cameras, satellites, or industrial sensors.

The quality of the captured image directly affects the performance of the entire Computer Vision system. Factors such as lighting conditions, camera resolution, image noise, viewing angle, and focus determine the usefulness of the captured data.

Common image formats include JPG, PNG, BMP, and TIFF. Video data is usually stored in formats such as MP4 or AVI.

2. Image Processing

Image processing improves image quality before analysis. Raw images often contain noise, poor lighting, or unnecessary details that can reduce detection accuracy.

Common preprocessing techniques include:

- Grayscale Conversion
- Gaussian Blur
- Median Blur
- Thresholding
- Edge Detection
- Histogram Equalization
- Morphological Operations

OpenCV is one of the most widely used libraries for performing image processing tasks because it provides efficient implementations of these algorithms.

3. Feature Extraction

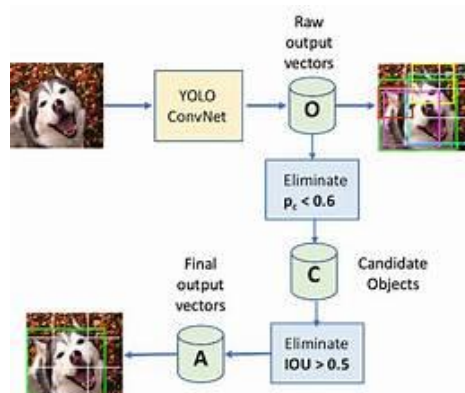
Feature extraction identifies important characteristics within an image that can help distinguish one object from another. Examples include edges, corners, shapes, textures, colors, and key points.

Traditional Computer Vision methods use algorithms such as SIFT, SURF, ORB, and HOG to extract features. These features are then used for object recognition or classification.

4. Object Detection

Object detection identifies the presence and location of one or more objects within an image. Unlike image classification, object detection predicts both the object category and its bounding box.

- YOLO (You Only Look Once)
- SSD (Single Shot Detector)
- Faster R-CNN
- RetinaNet



5. Image Classification

Image classification assigns a label to an entire image. A trained machine learning or deep learning model analyzes the image and predicts the most likely category.

Examples include identifying whether an image contains a cat, dog, flower, or vehicle. Deep learning models such as ResNet, EfficientNet, MobileNet, and Vision Transformers have significantly improved classification accuracy.

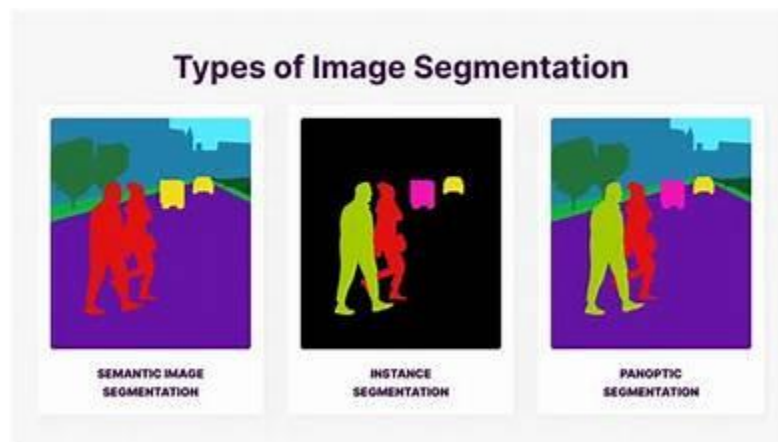
6. Image Segmentation

Image segmentation divides an image into multiple meaningful regions by assigning a class label to every pixel.

There are three major types of segmentation:

- Semantic Segmentation
- Instance Segmentation
- Panoptic Segmentation

Segmentation is commonly used in medical imaging, agriculture, satellite imagery, autonomous driving, and industrial inspection.



7. Object Tracking

Object tracking follows the movement of objects across consecutive video frames. It is commonly used after object detection in video applications.

Tracking algorithms estimate the object's position over time while handling changes in lighting, movement, and partial occlusion.

Applications include sports analytics, CCTV surveillance, autonomous vehicles, drone navigation, and human activity monitoring.

8. Deployment

After developing and testing a Computer Vision model, it is deployed for real-world use. Deployment allows users to interact with the application through desktop software, web applications, cloud services, or embedded devices.

Popular deployment platforms include Streamlit, Flask, FastAPI, Docker, AWS, Microsoft Azure, Google Cloud, NVIDIA Jetson, and Raspberry Pi.

Real-World Applications

Computer Vision is transforming many industries.

Healthcare:

Medical image analysis and disease diagnosis.

Agriculture:

Crop monitoring and disease detection.

Transportation:

Traffic monitoring and autonomous driving.

Retail:

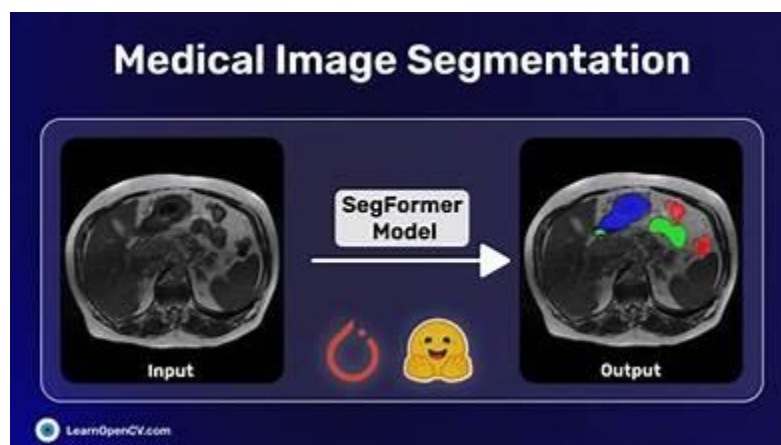
Automatic checkout systems and inventory management.

Manufacturing:

Quality inspection and defect detection.

Security:

Face recognition and surveillance systems.



Conclusion

The modern Computer Vision pipeline transforms raw visual information into meaningful insights through a sequence of carefully designed stages. Starting from image acquisition and preprocessing to feature extraction, object detection, classification, segmentation, tracking, and deployment, each stage contributes to the overall performance of the system. Understanding this pipeline provides a strong foundation for building advanced Computer Vision applications and prepares engineers for more complex deep learning tasks.

References

1. OpenCV Documentation
<https://opencv.org/>
2. Ultralytics YOLO Documentation
<https://docs.ultralytics.com/>
3. Python Official Documentation
<https://docs.python.org/3/>
4. OpenCV-Python Tutorials
<https://docs.opencv.org/>
5. Richard Szeliski, Computer Vision: Algorithms and Applications.